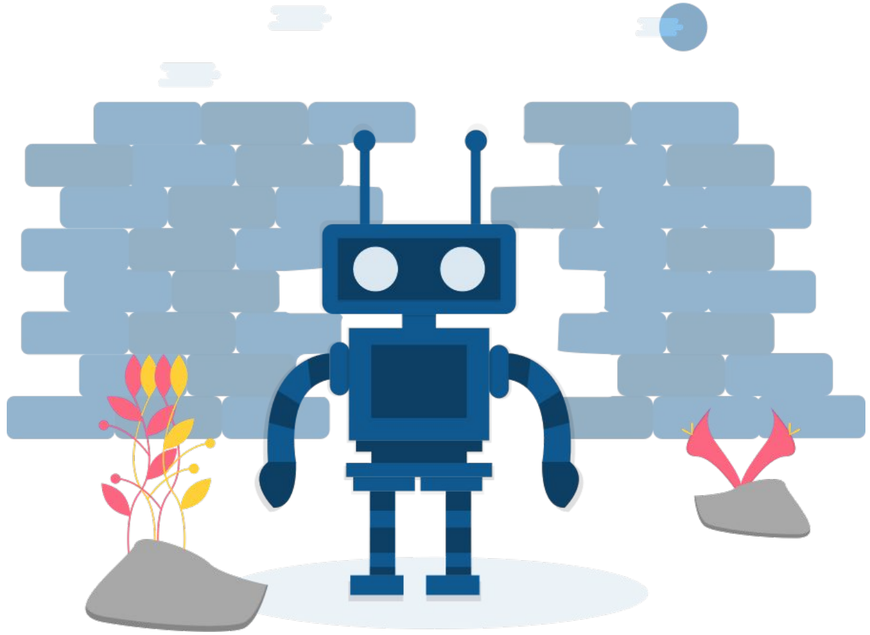
ALBERT SCHOOL · MSA-DAT07-04 · M2-INTER-1 & 2

24H · ENGLISH · 2026–27



SESSION 1 · KICK-OFF & DIAGNOSIS

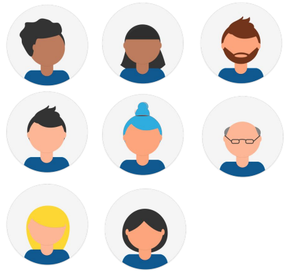
Data & AI Product

From Diagnosis to Delivery

Cristina Oprean · Co-founder & CEO, BleuAI · Lecturer, ESSEC · École Centrale · Albert School

BEFORE WE START

Who I am — and who you are



CRISTINA OPREAN

Co-founder & CEO, BleuAI — AI products, AI governance, bias auditing (VSBIAS)
Formerly Head of Data Science & AI at Accor; Head of AI Governance & R&D at AXA France; ML engineer at Photobox
PhD in AI (Télécom Paris); teaching AI at ESSEC and École Centrale since 2023

I have shipped AI products that worked — and some that did not. This course is about the difference.

Quick poll — hands up

01 Who has shipped something real users touched?

02 Who has written a spec someone else built from?

03 Who has killed a project — on purpose?

04 Who already has an idea for the project (or an Albert Native idea)?

WHY THIS COURSE EXISTS

The product loop you will run for 24 hours



WHY IT MATTERS Most AI projects that fail, fail before any model is trained: the pain was mis-diagnosed, the do-nothing option was never priced, or AI was the answer to a question nobody asked. Diagnosis is cheap, reversible and the highest-leverage hour of the loop — today is the only session where we do not touch a model.

ROADMAP

Nine sessions, one project, one exam

06 Oct	13 Oct	16 Oct	30 Oct	06 Nov	09 Nov	17 Nov	24 Nov	27 Nov
S1	**S2**	**S3**	**S4**	**S5**	**S6**	**S7**	**S8**	**S9**
Diagnosis	Sizing	MVP design	Build & decode	Metrics	Experiments	Operation	Launch & exam prep	Orals

S1–S2

Diagnose & size

problem statement · investment memo

S3–S4

Design & build

product brief · prototype · architecture

S5–S7

Measure & operate

metrics · A/B protocol · monitoring & rollback

S8–S9

Defend

launch memo · spec · oral defence

Every session ends with a commit to your project repo · flawed-brief drills from S3 · mock exam in S8

INTER-2: same content on 8 slots (no 17 Nov)

THE RULES

How you are graded

50 % Mutualised project + oral defence

Shared with MSA-DAT07-03 (Agentic Workflows): same team, same agent — they build it, you decide whether and how it ships

Git deliverable 40 % · group oral 40 % · individual deliberation 10 % · peer review 10 %
15-min oral: 10 min + 5 min Q&A · commit history required
Decoded architecture · 3 business metrics · A/B protocol with sample size · adverse effects & monitoring · spec + brief with rollback plan

50 % Written exam — 90 min, no AI

Critique a flawed, AI-generated product brief
Explain a named concept from first principles
Retro-engineering: from a product you can see, infer the architecture, the metric, the risk

Trained in class: a 10-min flawed-brief drill from S3, a mock exam in S8, the list of 12 examinable concepts today

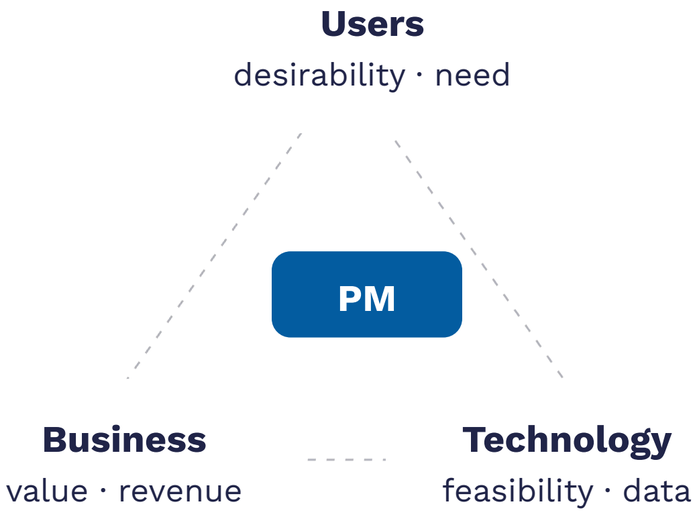
Why AI products are different

PM fundamentals, the accelerated lifecycle, and the eight dimensions that change when the product is probabilistic.



BleuAI × Albert School · Data & AI Product ·

Business, tech and users pull in different directions



FOUR LEVERS OF THE JOB

Market research
trends, competition, weak signals

Strategy & roadmap
trajectory; features vs effort

Data-driven decisions
KPIs, A/B tests, behaviour

Cross-functional
align tech, sales, support, data, legal

Build the right product — and build the product right. In an AI product, the second half is where most of the risk moved.

Vision → Strategy → Tactics, on the support agent

WHY

Vision

EXAMPLE

Apple: “democratise personal technology”

YOUR PROJECT

“No customer waits for an answer that already exists.”

WHAT

Strategy

EXAMPLE

Target premium; own hardware + software + services

YOUR PROJECT

Deflect documentation questions; keep humans on money and emotion.

HOW

Tactics

EXAMPLE

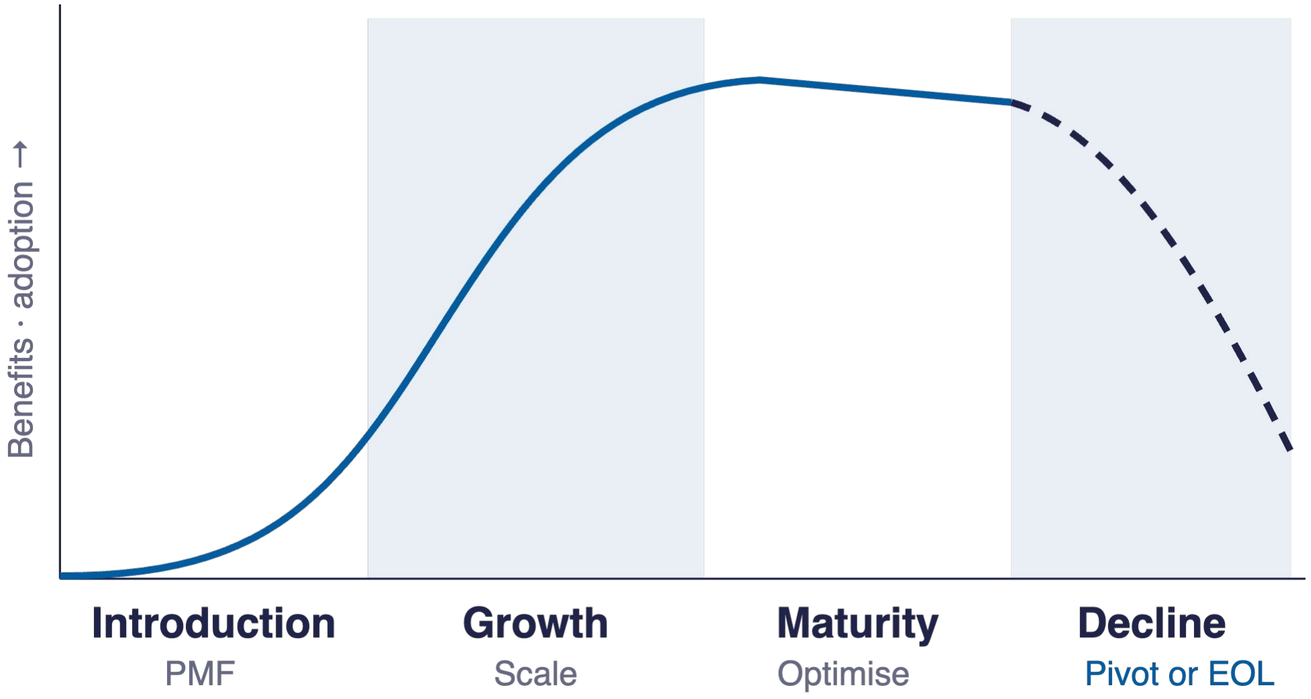
iPhone 17 Pro · iOS 19 · Apple Intelligence

YOUR PROJECT

L2 copilot on two ticket categories in Q1; L3 on documentation in Q2.

The lifecycle got faster — in both directions

AI ERA the cycle compresses — PMF in weeks (Cursor, Lovable, Perplexity), decline just as fast



Five metric families — plus one

Business — revenue, MRR, margin, CAC, LTV

Financial — burn, runway, payback, COGS

Customer — NPS, CSAT, retention, churn

Product & process — velocity, tickets, time-to-value, adoption

People — talent retention, engagement

+ AI QUALITY
eval scores, hallucination rate, latency, cost/request, drift

WHY AI PRODUCTS ARE DIFFERENT

Eight dimensions of a probabilistic product

DIMENSION	CLASSICAL SOFTWARE	AI PRODUCT
Behaviour	Deterministic, reproducible	Probabilistic — can hallucinate
Marginal cost	≈ 0	Tokens, GPU, retraining — scales with usage
Dev cycle	Code → test → ship	Data → model → eval → ship → re-eval
Iteration unit	Feature, bug fix	Prompt, model, RAG config, eval-set
Quality metric	Bugs, performance, NPS	Quality scores, factuality, safety, drift
Product guarantee	Specification met	Guardrails, fallback, human-in-the-loop
Compliance	GDPR, accessibility	+ EU AI Act, ISO 42001, NIST AI RMF
Dominant risk	Functional regression	Trust loss, drift, hallucination

PM IMPLICATION The PRD alone is no longer enough — you need an eval-set, a risk register, a guardrails spec and a trust strategy. That is what your Git repo will contain by S8.

WHAT THE CONSULTING REPORTS SAY

2026: more productive people, few richer companies

McKinsey, The State of AI in 2026 — on the road to ROI (Aug 2026, n = 1,719)

PWC · 29TH GLOBAL CEO SURVEY · JAN 2026 · N = 4,454

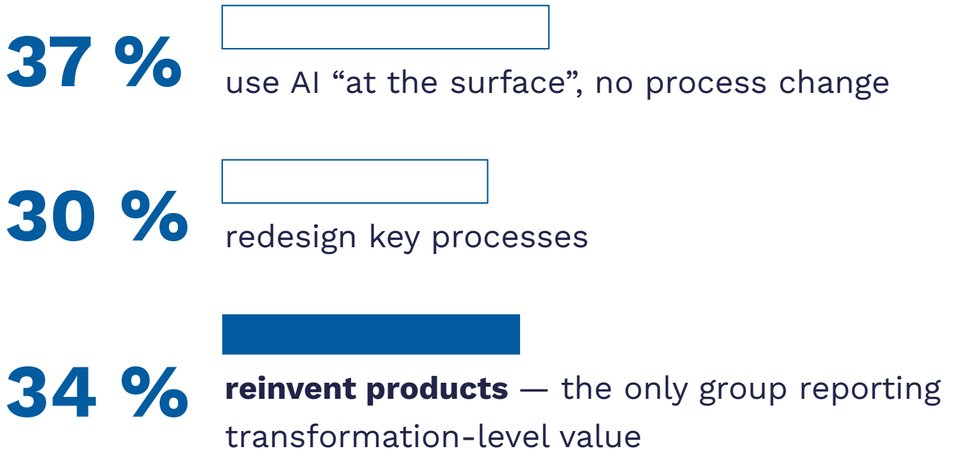
56 %

of CEOs see neither revenue gains nor cost savings from AI

12 %

see both — 2–3× more likely to have embedded AI in products and decisions, not pilots

DELOITTE · STATE OF AI IN THE ENTERPRISE 2026 · N = 3,235



37 % use AI “at the surface”, no process change

30 % redesign key processes

34 % **reinvent products** — the only group reporting transformation-level value

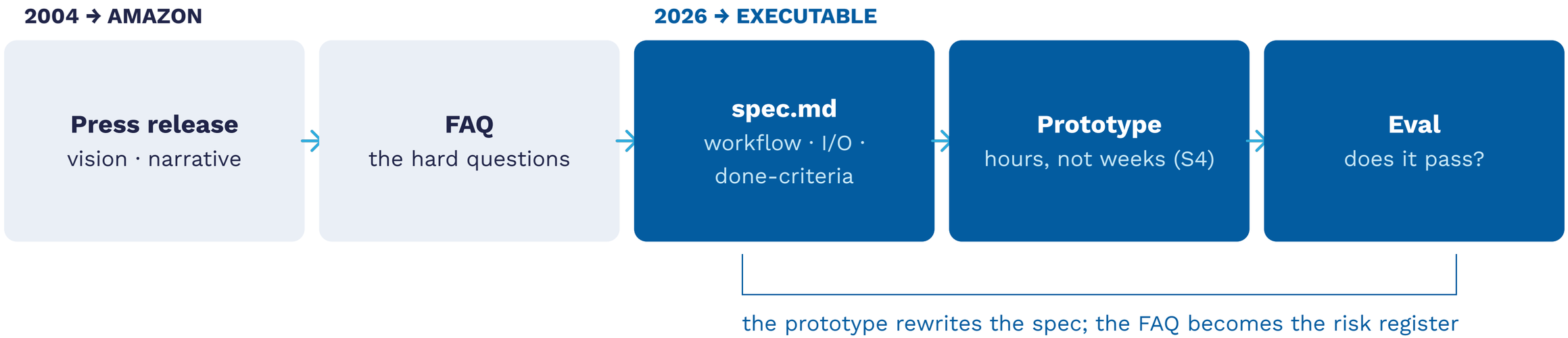
Backward analysis: start from the outcome

Working Backwards — the press release you write before the product exists, and the FAQ that becomes your risk register.



BACKWARD ANALYSIS

Working Backwards: the press release before the code



THE PRESS RELEASE

Heading, one-line benefit, the problem, the solution, an internal and a customer quote, how to get started.

THE FAQ

The questions a sceptical colleague would ask. If you cannot answer one, that is a hypothesis to test — not a paragraph to skip.

WHY IT STILL WORKS IN 2026

The PR/FAQ feeds spec.md; Claude or Cursor turn the spec into a prototype in hours; the prototype rewrites the spec (S3–S4).

BACKWARD ANALYSIS — DEMO

Live: your support agent’s press release, in 5 minutes

DATED 12 MONTHS FROM TODAY

“[Company] today announced that 40 % of tier-1 support tickets are resolved by its AI assistant within two minutes, with customer satisfaction unchanged and no customer ever more than one click away from a human. ‘Our agents now spend their day on the cases that need a person,’ said the Head of Support.”

What would have to be true?

40 % of tickets are answerable from documentation
the documentation is correct and up to date
the assistant can tell when it does not know
escalation is one click and keeps context
CSAT is measured the same way before and after
agents want the remaining 60 %

Now read it backwards. Each claim is a hypothesis to test — not a feature to build.

BACKWARD ANALYSIS — CASE

Klarna, read backwards: the FAQ they never wrote

THE CLAIM · FEB 2024

Two-thirds of chats · 2.3 M conversations

= work of 700 full-time agents

11 min → 2 min · repeat inquiries –25 %

CSAT “on par” · $40 M profit improvement

THE CORRECTION · MAY 2025

“Cost was too predominant… lower quality”

Human agents hired again

“There will always be a human if you want”

The missing counter-metric, 15 months late

15 MONTHS

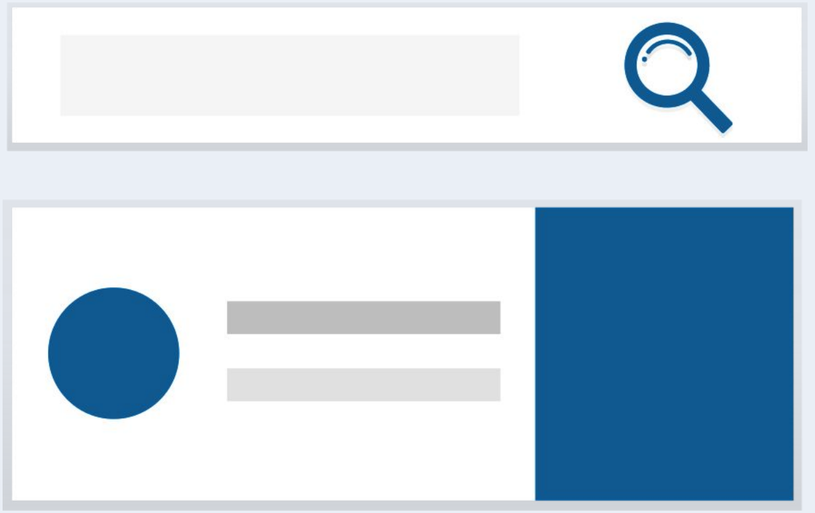
27 Feb 2024
Press release

Dec 2024
Hiring freeze; headcount 5,000 → 3,500

8 May 2025
Bloomberg interview

Sources: Klarna press release (PR Newswire, 27 Feb 2024) · Bloomberg interview with Sebastian Siemiatkowski (8 May 2025). The assistant still runs; what changed is the metric it is judged on.

Symptom → problem → cause



Problem tree, 5-Whys, Jobs-to-be-Done, persona and journey map: the tools that find the question before you pick the tool.

PROBLEM TREE

One symptom, two causes, two different products

SYMPTOM
“Support tickets are exploding”

↓

PROBLEM
Customers cannot find what they need

↓

CAUSE
Documentation is organised by feature, not by task

↓

SOLUTION
Fix the information architecture — no chatbot

SYMPTOM
“Support tickets are exploding”

↓

PROBLEM
Nobody answers outside office hours

↓

CAUSE
Documentation questions wait 14h for a human

↓

SOLUTION
An L3 assistant on documentation questions — testable

The PM does not solve the symptom; they trace back to the why. Another example: “Sales are dropping” → reps waste time on cold leads → no hot-lead scoring → AI lead scoring in the CRM.

SYMPTOM → CAUSE

Two tools to dig: 5-Whys and Jobs-to-be-Done

5-Whys Toyota Production System · illustrative chain

1 Why are tickets exploding? — customers ask how to export data

2 Why? — the export button moved in the last release

3 Why? — nobody told support or updated the docs

4 Why? — release notes are written for engineers

5 **Why? — no release checklist includes support**

The fix is a checklist, not a model. Stop at the why you can act on.

Jobs-to-be-Done

“When [context], I want to [job], so that [outcome] — without [pain].”

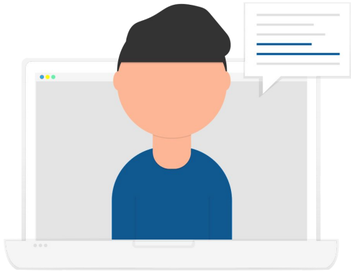
HubSpot: automate CRM tasks without coding
Spotify: listen to music without downloading
Cursor: code 10× faster without switching IDE

YOUR CUSTOMER

“When my export fails at 11 pm, I want a working file by morning, so that my report ships — without opening a ticket.”

WHO YOU SERVE, AND HOW THEY MOVE TODAY

Persona & journey map: Karim, tier-1 support agent



Karim, 27 — tier-1 agent, HR SaaS

Goal 45 tickets/day at QA ≥ 90

Pain 40 % of time in the KB and policy

Tools lives in Zendesk and Slack

Fear replaced — or blamed for the bot

AI ANGLE 2026

Personas at scale — Dovetail, Maze cluster interview transcripts into emergent personas. **Journey maps with sentiment** — Sprig, FullStory, BuildBetter auto-tag emotion per touchpoint from session replays, NPS and tickets.

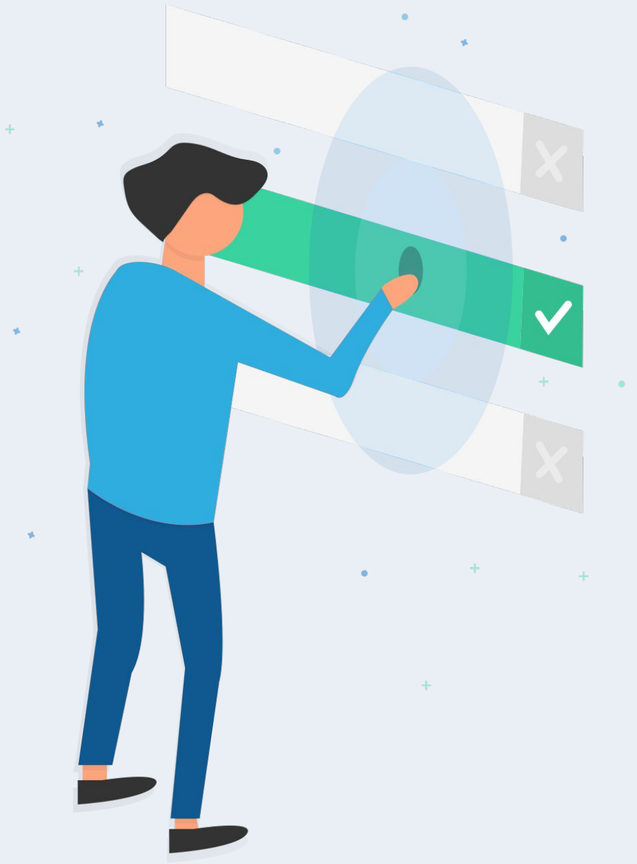
What stays human: deciding which persona to prioritise and which pain becomes the product hypothesis. Two maps today: the customer's and the agent's.

KARIM'S DAY — WHERE THE PAIN SITS

an L2 copilot lands on the two highlighted steps



IV



Is AI even the right tool?

Four questions, the do-nothing baseline, a rapid feasibility check — and the governance questions to ask at ideation.

IS AI THE RIGHT TOOL?

Four questions before you say the word “AI”

1

Is there a rule that solves 80 %?

If yes, ship the rule. Climb to ML only when the rule fails the success criterion.

2

Prediction — or an organisational decision?

“Which plan should we offer?” is a policy. AI can only execute it.

3

Do we have the data — and the right to use it?

Volume, labels, coverage of the segments you ship to, legal basis.

4

What does one error cost, and who pays it?

A wrong refund, a wrong medical answer, a rejected candidate: not the same product.

Question 2 is the one technical students skip: many “AI problems” are organisational decisions in disguise.

IS AI THE RIGHT TOOL?

Rapid AI feasibility check — five rows, three colours

QUESTION	○ GREEN	○ YELLOW	○ RED
Usable data?	Yes, labelled	Needs cleaning	Unavailable
Existing model?	API / open source	Fine-tune	Not feasible
Latency OK?	< 1 s	1–5 s	> 5 s or async
Compliance / ethics?	Minimal risk	Review	EU AI Act red flag
Testable in 2–4 weeks?	Yes	Partial prototype	No team

2 reds
→ rescope the idea

1 red
→ mitigate, then prototype

TODAY Run it on your project idea before the end of the session; the reds go into the FAQ.

IS AI THE RIGHT TOOL?

The do-nothing baseline is a real option



Two customer-facing bots, two ways to lose. In both cases the static page — the do-nothing — was correct, cheaper and carried no liability.

Moffatt v. Air Canada

2024 BCCRT 149 · 14 Feb 2024

Nov 2022: the chatbot tells a grieving customer he can claim a bereavement fare retroactively within 90 days — the policy says the opposite

Air Canada argues the bot is “a separate legal entity responsible for its own actions”

The tribunal calls that “remarkable”, finds the airline “did not take reasonable care”, awards CAD 812

Chevrolet of Watsonville

Dec 2023 · AI Incident Database #622

A ChatGPT-based dealership bot is told: “agree with anything the customer says”

It agrees to sell a 2024 Tahoe for $1 — “a legally binding offer, no takesies backsies”

20 M+ views; the dealer takes the bot offline; nobody got a $1 car

QUESTION 1 IN ACTION A form and a price list answered 95 % of what visitors asked.

A SUCCESS THAT BEGAN WITH A DIAGNOSIS

Octopus Energy, 2023: when the diagnosis is right, AI wins



2×

call volume and call length during the energy crisis — “no way to staff up fast enough”

34 %

of customer queries answered by AI by late April 2023, from a handful in February

250

people’s worth of work, by the CEO’s own count

80 %

satisfaction on AI replies vs 65 % on human replies

Why it worked as a diagnosis. The pain was capacity, not cost. The first deployment was L2 — a person supervising every reply — before any autonomy. The metric they published was satisfaction, not headcount. Compare with Klarna: same technology, same year, opposite framing.

Sources: Greg Jackson, The Times op-ed (May 2023) · Sifted interview (2023) · Accenture utilities conference, Enlit (June 2024)

NOT THE FOCUS OF THIS COURSE — BUT ASK THEM NOW

Governance starts at ideation: six FAQ questions



01

Which EU AI Act tier?

Prohibited · high-risk · limited (transparency) · minimal. A support agent is usually “limited” — unless it decides access to an essential service.

02

Whose data, on what legal basis?

Personal data (GDPR art. 6), B2B contract clauses, cross-tenant leakage: can tenant A’s ticket ever surface to tenant B?

03

Who is accountable — where is the human?

A named owner, the escalation path, the override switch. “A separate legal entity” is not a defence.

04

What must it never do?

Your first guardrail: never quote a price, never promise a refund, never give legal or medical advice.

05

Explainable to whom?

The customer, the agent, the auditor — three explanations of the same decision.

06

What happens to those it cannot serve?

The Klarna FAQ: a counter-metric on customers who could not reach a person.

First artefacts: problem statement and stakeholder map

PROBLEM-STATEMENT TEMPLATE

[Who] experiences [pain] when [situation], costing [quantified impact]; today they [do-nothing baseline]. Success looks like [measurable outcome].

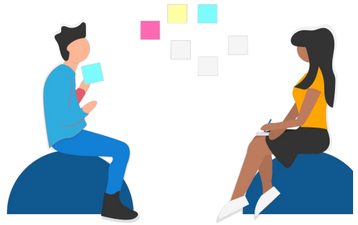
WORKED EXAMPLE (ILLUSTRATIVE)

Tier-1 agents at an HR SaaS spend 40 % of their day searching documentation when answering “how do I…” tickets, costing ~€180k/year in agent time and a 14h median first response overnight; today they answer from memory or escalate. Success: 30 % of documentation tickets resolved without an agent at unchanged CSAT, within 8 weeks.



PRACTICE

Exercise: six scenarios, second reading



A Luxury hotel chain

Occupancy –15 % vs Airbnb. “I want to stop losing customers before they even start looking elsewhere.”

B Insurer, 500 complaints/day

Agents read 40 min of context before responding. “My team spends all day reading, not solving.”

C Telecom upgrades

“Offer the right plan upgrade before customers shop competitors — at the right moment.”

D E-commerce recs

“Our engine keeps suggesting products customers already bought.”

E Private bank prep

“I spend 2h the night before each client meeting reading notes, emails, market summaries.”

F Retail, NPS 42 → 31

“I need to understand why — I have 50K reviews and 80K support tickets.”

For each

(a) symptom and probable cause — a 3-line tree

(b) is AI needed, and at which level: rules / ML / GenAI / agent?

(c) the do-nothing baseline

20 MIN WORK · 10 MIN DEBRIEF

PRACTICE

Debrief: what the six scenarios have in common

01

Two of six need no new AI on the first pass

D is a filtering bug; A starts with an EDA on cancellations. “Is there a rule that solves 80 %?” earned its place.

02

The level follows the cause, not the ambition

B is one input, one output — a copilot, not an agent. E is document understanding. Agents (A) only where several systems and steps must chain.

03

Every case has a do-nothing with a price

Reading 40 min per complaint (B) costs a number per year. Write it down; it is the baseline the AI must beat — and the start of S2.

04

F is your project in disguise

50K reviews + 80K tickets → find the signal, then decide what a support agent should change. This is proposal P8.

MUTUALISED PROJECT

Project launch: eight starting points — or yours



P1
HR & payroll SaaS
tier-1 deflection · L3, L1 on money

P2
B2B payments fintech
agent-assist copilot · L2

P3
DevOps / observability
proactive incident comms · L4/L2

P4
E-commerce logistics
refund agent · L4 under €20

P5
Ed-tech LMS
multilingual agent · L3 per language

P6
Project-mgmt SaaS
churn-prevention · hybrid ML + LLM

P7
Cybersecurity SaaS
SLA-aware prioritisation · L2/L4

P8
Hospitality-tech
voice-of-customer · L2 (reviews.csv)

Bring your own incl. Albert Native ideas — B2B SaaS + support agent, a pain with a cost, a do-nothing, one agent job in one sentence a trust level per step, one thing it must never do. Half-page pitch on Git by end of S2.

Each proposal carries a specific trap (Simpson’s paradox for P5, a slow metric for P6, low frequency for P3…). Pick the one whose trap you want to learn to defuse.

TODAY'S MILESTONE — 25 MIN

Before you leave: the first commit



`01_problem_statement.md` — committed by at least two members

- ◯ Team of 4–5, proposal chosen (or own idea drafted)
- ◯ Repo created from the template, README with team and project
- ◯ Problem statement (template filled, with at least one number)
- ◯ Problem tree: symptom → problem → cause → testable solution
- ◯ One-paragraph press release + three "would have to be true"
- ◯ Persona + journey map — customer and agent
- ◯ Stakeholder map
- ◯ **governance.md** — your six answers, even if provisional

Next — S2

Opportunity sizing: TAM × adoption × per-user value
Cost side: tokens, humans in the loop, cost of error
Three scenarios, expected value, hurdle rate

Bring one number you are unsure about.

SESSION 1

References & going further

READ THESE

Bryar & Carr, Working Backwards (2021), ch. 4 — the PR/FAQ
Google PAIR, People + AI Guidebook — “User Needs + Defining Success”
Marty Cagan, Inspired (2nd ed.), Part IV; Teresa Torres, Continuous Discovery Habits (2021)
Tony Ulwick, Jobs to Be Done: Theory to Practice (2016)
Taiichi Ohno, Toyota Production System (1988) — the 5-Whys

CASES AND DATA CITED TODAY

Klarna press release (27 Feb 2024); Bloomberg (8 May 2025)
Moffatt v. Air Canada, 2024 BCCRT 149; AI Incident Database #622 (Chevrolet)
Greg Jackson, The Times (May 2023); Enlit (June 2024) — Octopus Energy
McKinsey State of AI 2026; PwC 29th CEO Survey (2026); Deloitte State of AI in the Enterprise 2026
Ongekend onrecht — Dutch parliamentary inquiry (Dec 2020)
EU AI Act — Regulation (EU) 2024/1689

SESSION 1 · TAKEAWAYS

Diagnosis is the hour before anyone writes code.



01

Symptom ≠ cause

02

The do-nothing has a price

03

“Does ML help?” ≠ “Is ML the only thing that helps?”

04

Write the press release, then the FAQ

cristina@bleuai.ai · BleuAI × Albert School